Passage-1

The emissions humans put into the atmosphere now will affect the climate in the middle of the century and onwards. Technological change, meanwhile, could make a future transition away from fossil fuels cheap or it might not, leaving the world with a terrible choice between sharply reducing emissions at huge cost or suffering through the effects of unabated warming. Businesses that do not hedge against the threat of uncertain outcomes fail. The world cannot afford such recklessness on climate change.

11. Which one of the following statements best reflects the crucial message conveyed by the author of the passage?

(a) Businesses that cause emissions may need to close down or pay for pollution in future.
(b) The only solution is technological development related to the issues of climate change.
(c) Waiting to deal with carbon emission until technology improves is not a wise strategy.
(d) Since future technological change is uncertain, new industries should be based on renewable energy sources.

Ans. (c) : According to the passage, statement that best reflects the crucial message conveyed by the author is - "Waiting to deal with carbon emissions until technology improves is not a wise strategy".

Passage-2

Environmental problems cause health problems. Substantial changes in lifestyle can reduce environmental or health problems, but this idea appears almost impossible to adopt with environmental problems, individual efforts can be perceived as having a negligible effect and therefore lead to inertia. With health, on the other hand, individual choices can make the difference between life and death, literally. And yet, barring a few, there seems to be the same collective lethargy towards making their choices.

12. Which one of the following statements best implies the most rational assumption that can be made from the passage?

(a) We are likely to spend more money on cure than prevention.
(b) It is the job of the government to solve our environmental and public health problems.
(c) Health can be protected even if environmental problems go on unattended.
(d) Loss of traditional lifestyle and the influence of western values led to some unhealthy ways of living.

Ans. (c) : Statement that best implies most rational assumption that can be made from the passage is- "Health can be protected even if environmental problems go on unattended".

Passage-3

Many people are not eating the right food. For some, it is simply a decision to stick with food they enjoy but which is not too health. This is leading to an increase in non-communicable diseases. This in turn leads to major burden on our health-care systems that have the potential to derail the economic progress which is essential for the poor to improve their lives. For others, it is about limited access to nutritious food or a lack of affordability, leading to monotonous diets that do not provide the daily nutrients for them to develop fully. Part of the reason nutrition is under threat worldwide is that our food systems are not properly responding to nutritional needs. Somewhere along that long road from farm to fork, there are serious detours taking place.

13. Which one of the following statements best reflects the crux of the passage?

(a) The scheme of Universal basic Income should be implemented worldwide as a way of poverty alleviation.
(b) We must place food-based nutrition at the centre of our policy debate.
(c) Nutritional status of food should be improved by creating appropriate genetically modified crops.
(d) Using modern food processing technologies, we must fortify food items with required nutrient elements.

Ans. (b) : Statement that best reflects the curse of the passage is- "We must place food based nutrition at the center of our policy debate".

14. Three of the five positive integers p, q, r, s, t are even and two of them are odd (not necessarily in order). Consider the following:
1. p, q, r, s, t is definitely even.
2. 2p + q + 2r – 2s + t is definitely odd.
Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : Let p, q, r are even numbers and s, t are odd numbers
Statement 1 - (p + q + r) – (s + t)
We know, sum of three even numbers is a even number, therefore, p + q + r = even
And sum of two odd numbers is even therefore, s + t = even
even – even = even
i.e. (p + q + r) – (s + t) = even number
Statement 1 is correct.
Statement 2 - (2p + q + 2r – 2s + t) cannot be always odd.
Hence, this is wrong statement.

15. Consider the following in respect of prime number p and composite number c.

1. $\frac{p+c}{p-c}$ can be even.
2. p + c can be odd
3. pc can be odd
Which of the statements given above are corret?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d) : Let p = 11 and c = 9
Statement 1 - $\frac{p+c}{p-c}$
$= \frac{11+9}{11-9} = \frac{20}{2} = 10 = \text{even}$
Statement 1 is correct
Statement 2 - $= 2 \times p + c$
$= 2 \times 11 + 9 = 22 + 9$
$= 31 = \text{odd}$
Statement 2 is correct
Statement 3 - pc
$= 11 \times 9$
$= 99 = \text{odd}$
Statement 3 is correct.

16. A 3-digit number ABC, on multiplication with D gives 37DD where A, B, C and D are different non-zero digits. What is the value of A + B + C?

(a) 18
(b) 16
(c) 15
(d) Cannot be determined due to insufficient data

Ans. (d) : According to the given,
D (100 A + 10B + C) = 3000 + 700 + 10D + D
= 3700 + 11D
Case (i) - Let D = 4
The number 3700 + 11D = 3744
$100\,A + 10B + C = \frac{3744}{2} = 936$
Therefore, A = 9, B = 3 and C = 6
Case (ii) - Let D = 5
The number (3700 + 11D) = 3755
$100A + 10B + C = \frac{3755}{5} = 751$
Therefore, A = 7, B = 5, C = 1
There is no unique value for (A + B + C)
Hence, A + B + C cannot be determined

17. For any choices of values of X, Y and Z, the 6-digit number of the form XYZXYZ, is divisible by :

(a) 7 and 11 only (b) 11 and 13 only
(c) 7 and 13 only (d) 7, 11 and 13

Ans. (d) : According to the question,
Any number of the form 'XYZXYZ' will be a multiple of 1001
Then, irrespective of the values of X, Y and Z, this 6 digit number will be divided by 1001
On factorizing 1001, we get, 1001 = 7 × 11 × 3
Hence, 'XYZXYZ' is divisible by 7, 11 and 13.

18. 125 identical cubes are arranged in the from of a cubical block. How many cubes are surrounded by other cubes from each side?

(a) 27 (b) 25
(c) 21 (d) 18

Ans. (a) : According to the question.
→ If 125 identical cubes are arranged in the form of a cubical, then, there will be 25 cubes in each of the 5 layers.
→ Now, separating the upper layer and lower layer total number of cubes separated = 50
→ Now, separating the outer 16 cubes from the remaining three rows, number of removed cubes = 16×3 = 48
→ Total number of cubes which are not surrounded by other cubes from each side = 50 + 48 = 98
→ Therefore, number of cubes surrounded by other cubes from each side = 125 – 98 = 27

19. How many distinct 8-digit numbers can be formed by rearranging the digits of the number 11223344 such that odd digits occupy odd positions and even digits occupy even positions?

(a) 12 (b) 18
(c) 36 (d) 72

Ans. (c) : According to the question,
In an eight digit number, even positions will be 2, 4, 6, 8 and odd positions will be 1, 3, 5, 7.
There are 4 even digits and they will occupy even positions, so they can be arranged is 4!.
But is these 4 digits, two digits are repeated, so now these 4 even digits are arranged in $\frac{4!}{2! \times 2!} = 6$
Similarly, 4 odd digits are arranged in $\frac{4!}{2! \times 2!} = 6$
Therefore, the odd and even digits that can be arranged as per the given condition will be 6×6 = 36.

20. A, B, C working independently can do a piece of work in 8, 16 and 12 days respectively. A along works on Monday, B alone works on Tuesday, C alone works on Wednesday; A alone, again works on Thursday and so on.
Consider the following statements:
1. The work will be finished on Thursday.
2. The work will be finished in 10 days.
Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : A alone can do a piece of work in 8 days
B alone can do a piece of work in 16 days
C alone can do a piece of work in 12 days
LCM of 8, 16 and 12 = 48
Let total work is 48 units
Then,

Monday	-	6 unit	
Tuesday	-	3 unit	
Wednesday	-	4 unit	
Thursday	-	6 unit	
Friday	-	3 unit	
Saturday	-	4 unit	
Sunday	-	6 unit	
Monday	-	3 unit	
Tuesday		-	4 unit
Wednesday	-	6 unit	
Thursday	-	3 unit	
		48 unit	

The work will be finished on Thursday.

Passage-1

To tackle the problem of pollution in cities, policy makers think that drastic actions like temporary use of odd-even number scheme for vehicles, closing schools, factories, construction activates, and banning the use of certain type of vehicles are a way forward. Even then the air is not clean. Vehicles more than 15 years old comprise one percent of the total; and taking them off the road will not make any difference. Banning certain fuels and car types arbitrarily is not proper. Diesel engines produce more PM 2.5 and less CO_2 than petrol or CNG engines. On the other hand, both diesel and CNG engines produce more NO_x than petrol engines. No one has measured the amount of NO_x that CNG engines are emitting. Arbitrary bans on vehicles that have passed mandated fitness tests and periodic pollution tests are